Food Shelf Life Control

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Food Shelf Life Control (A14863)

Short Title: Food Shelf Life Control
Faculty Science and Computing
Department Science
Cluster Food Science
Author ROMAHONY
Credits 5 **Level:** Advanced

Description of Module / Aims

The aim of this module is to cover the theoretical aspects surrounding the shelf life of food. Factors that influence shelf life along with methods for extending the shelf life of various foods are visited. Analysing and predicting the shelf life of food is performed on a practical basis in this module. Predicting the rates of shelf life using modern day technology is investigated.

Programmes

FOOD-0005	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	4 / 8 / M
FOOD-0005	BSc (Hons) in Food Science and Innovation (WD_SFDSC_B)	4 / 2 / M

Indicative Content

- Definitions around shelf-life, factors influencing shelf-life, types of deterioration
- Measuring and predicting shelf-life
- Design of shelf-life experiments
- Extending shelf-life
- Glass transition
- Modelling shelf-life: software models
- Accelerated shelf-life tests
- Advanced instrumental methods for starch retrogradation

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Establish the factors that affect the shelf-life of food.
2. Verify and critique how to measure, predict and extend the shelf-life of various foods.
3. Determine the microbial load and estimate shelf-life of a food product.
4. Evaluate various predictive computer modelling packages and relevant scientific literature.
5. Compose a laboratory-based group project to assess the microbial load of a specific food type.

Learning and Teaching Methods

- Blended learning approach.
- Pre-lecture activites include online discussion forums, use of online software programmes.
- Post-lecture activities include production of mindmaps as a method of reviewing relevant scientific literature.
- Peer assessment for learning - students grade each other's work.
- Construction of flowcharts/portfolios to capture student learning activities within the laboratory setting.
- Discussion forums designed so that students can reflect on their contributions thus promoting reflective practice.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Group Project	40%	3,5
In-Class Assessment	30%	1,2
Assignment	30%	4

Assessment Criteria

<40%: Very little knowledge and understanding of the subject matter. Unable to carry out critical thinking or practical skills.

40%-49%: Demonstrate a limited knowledge of the subject matter. May have some problems with critical thinking and problem solving. Adequate practical skills.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Good practical skills.

60%-69%: Demonstrate skilled and sound knowledge of the subject matter. Show ability to analyse and logically discuss in an effective way. Good critical thinking and a high level of competence and efficiency in theoretical and practical areas of this subject.

70%-100%: All of the above to an excellent level. Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Supplementary Material

- Kilcast, D. and P. Subramaniam. _The stability and shelf life of food_. 2nd ed. Cambridge: Woodhead publishing limited, 2010.

Requested Resources

- COMPUTER LAB: Networks Lab
- Room Type: Biology Lab
- Lecture Room: Loose Seated